

Book Title: **Science and Sikhism: Conflict or Coherence**

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Dr. DP Singh is a prolific writer in many areas of Science, Religion and Literature. He came into my contact almost four decades back when he started his teaching career in Shivalik College, Nangal. In my note published on the blurb of this book, I wrote: " I expect his forthcoming book *"Science and Sikhism: Conflict or Coherence"* will prove to be a landmark in the area of Science-Religion Dialogue, with special reference to Sikh religion". I can declare without an iota of doubt that DP has more than justified my claim on his book.

The title of the book is followed by a note *"Anthology of Essays on various Concepts in Sri Guru Granth Sahib"* which may be considered as a sub-title to the book. Nearly a dozen titles have appeared in the global market projecting scientific vision of Sikh Gurus enshrined in the holy pages of Sri Guru Granth Sahib but there is no rival to this volume in my view till date. The coverage of topics by DP Singh is very exhaustive. All Chapters of the book open with an appropriate quotation from Science, Religion or Literature.

The book is dedicated to "Inquisitive Human Spirit" which is displayed by the author throughout this volume. The dedication is followed by the famous quote of Freeman Dyson, the renowned Physicist: *"Science and religion are two windows that people look through, trying to understand the big universe outside, trying to understand why we are here. The two windows give different views, but they look out at the same universe. Both views are one-sided, neither is complete. Both leave out essential features of the real world. And both are worthy of respect"*. Surprisingly, Dyson is contradicted as SGGS and Science are compatible in their world-view and there is no conflict but harmony between these two windows.

The preface *"A Spiritual Journey over a Scientific Bridge"* has been written by Solomon Naz, an ex-Professor of Comparative Religious Studies of York University, Canada. I agree with his remarks: "Dr. Devinder Pal Singh is one of those physicists who reinterpret religion under prismatic analysis of scientific knowledge. One great asset that I find in Dr. Devinder Pal Singh is his dedication to explain and expound the relationship of Bani (spiritual hymns), not only to the divine life alone, but the unbreakable and indivisible unity of spirit and

physicality. Unity in diversity is the central theme of Guru Granth Sahib and Dr. Devinder Pal Singh has explored all its aspects thoroughly".

The contents of this volume are divided into 21 Chapters for the sake of classification of topics under discussion. Chapter 1 begins with the famous quote of Albert Einstein: "*Science without religion is lame, religion without science is blind*". The author goes on to discuss the relationship of religion and science with mankind: "The interaction between science and religion has been quite fascinating. It has led to a new awakening, causing a perceptible change in their inter-relationship. With passage of time, accepting the scientific truths, religious thoughts undertook the process of replacement of the irrational beliefs with truth and rationality. In contrast, Sikhism is relatively a new religion. The ideas as expressed in its holy Granth – Sri Guru Granth Sahib (SGGS) – have been found to be more in tune with the modern scientific facts".

In Chapter 2, author calls SGGS as a unique religio-philosophic treatise and elaborates its universal perspective as follows: "The beauty and splendour of SGGS is that the revealed *Shabad* and reason are consistent. There are no dogmas; only universal and eternal rational beliefs and practices. The ideals, as laid down in SGGS, are pragmatic, logical and viable. These are perennial source of freedom and creativity, unique in the present-day world". The universal character of SGGS is fortified by quotes from *Gurbani* which makes it as a worthy tool for forging unity in mankind. According to author: "SGGS rejects all ritualism, formalism and symbolism. It defines a religion consisting of loving conviction and good will for fellow human beings. With such a universal approach, *Gurbani* is aiming at forging common bonds in the human race without discrimination of caste, creed, sex or nationality".

In 3rd Chapter "Guru Nanak's Teachings: A Scientific Perspective", author defines scientific perspective and tries to establish that Guru Nanak followed it in his life and teachings: "Guru Nanak's life was a continuous process of scientific experimentation and statement. Guru Nanak's life and writings are rich in several such examples wherein his scientific approach to resolve several real life situations with logic and rationality is evident". This unique character of founder of Sikh religion is brought into limelight so beautifully that it needs to be adopted by Sikh preachers (*Kathakars*) in their exegesis of *Gurbani*.

The following seven Chapters (4-10) starting with "Sunn (Primal Void): Nothing or Everything?" and ending with "Moon: An Esoteric Neighbour" may be considered as the

Core Chapters of this volume. Stephen Hawking in his last book denies the existence of God but believes in creation of Universe out of 'Nothing': "*Science predicts that many different kinds of universe will be spontaneously created out of nothing. It is a matter of chance which we are in*". However, concept of "Sunn (Primal Void)" has been advocated by Eastern religions, including Buddhism and Sikhism. The author has discussed this concept at length with reference to holy texts of Eastern religious traditions. He has invoked the Quantum field theory to establish the link between Modern Physics and "Sunn (Primal Void)".

The author has quoted profusely from the *Maru Solhe Raaga* and other Banis of SGGS to establish his thesis of 'Sunn':

jlu Qlu Drix ggnu qh nwhl Awpy Awpu klAw krqwr]
nw qid mwieAw mgnu n CwieAw nw sUrj cµd n joiq Apwr]
srb id®sit locn AB AMqir eykw ndir su iq®Bvx swr]

*In that state, there is no water, land, earth or sky; only the
Creator Lord Himself exists. There is no intoxication of Maya
there, and no shadow, nor the infinite light of the sun or the
moon. The eyes within the mind which see everything, with one
glance, they see the three worlds.*

(Gujri, M. I, SGGS, p. 503)

According to author: "The Sikh scripture describe the ultimate reality as Sunn – the *primal void* – a primal manifestation of the creative force (Creator) of the worldly phenomena. *Sunn* is not nothingness but represents a pregnant emptiness".

Chapter 5 "Creation of Universe: A Fabulous Tale" tries to establish the Sikh Cosmology but the author fails to refer to the work already published by his predecessors in this domain. There is nothing new at the conceptual level which has been discovered by the author. The most popular quote from the *Maru Solhe Raaga* has been repeated by the author:

Arbd nrbd DuµDUkwrw] Drix n ggnw hukmu Apwrw]
nw idnu rYin n cµdu n sUrju suµn smwiD lgwiedw]
*For endless eons, there was only utter darkness. There was no
earth or sky; there was only the infinite Command of the
Creator. There was no day or night, no moon or sun; The
Creator sat in primal, profound Samadhi.*

(Maru M. I, SGGS, p 1035)

I consider the Chapter "Panch Tattva: The Five Classical Elements" as one of the best contribution of author to the Sikh literature. He has discussed the concept of "Panch Tattva"

in the most elaborate manner and discovered its origin from the treasure troves of cultures of both East and West. I find a new interpretation of Japuji Sloka by the author: "Guru Nanak has emphasized the vital role of *panch tattva* (air, water, earth, day representing the fire element-sun and night representing the element akasha, with its numerous stars or worlds) for the growth and well-being of the whole world. Thereby Gurbani proclaims the importance of living in harmony with *panch tattvas*".

Chapter "Kudrat (Nature) in Guru Nanak's Holistic Vision" is also a masterpiece created by the author. As in Chapter 5, the omission of references to work of predecessors is an irritant for the reviewer. The author fails to record why Guru Nanak chose Arabic term 'Kudrat' to represent 'Nature', in preference to the Indian term 'Prakirti' used in Sankhya system. He writes that both these terms are synonymous (which is not true): "Guru Nanak has delineated in great detail about *nature* in his hymns. In his hymns, the word '*kudrat*' has often been used to depict 'creation' or *nature*. In Guru Nanak's *Bani*, the word '*kudrat*' is also synonymous with 'prakirti' or 'maya'". It clearly shows that he had overlooked the work of Sirdar Kapur Singh referred to by the reviewer in his essay "Concept of Nature (Qudrat) in Science and Sri Guru Granth Sahib".

The next three Chapters are focussed on Sun, Earth and Moon. The author has dug up cultural roots of origin of these heavenly bodies as they have been used as 'Metaphors' in Indian and other cultures. For example, earth has been referred to as 'Mother' in the Indian culture: "The earth, in a very real sense, is our mother. We are born from this mother, we are extensions of the earth and the cosmos of which it is a part. This means that our conceptualizing and our spirituality also extend from the spiritual dimension of the cosmos and the earth". Sikh Gurus have emphasized that the Creator is all pervading in His creation. The following hymns describe that the Creator is pervading in earth as well:

jil Qil mhlAil soie]

He (The Creator) is pervading the water, the land and the sky.

(M. V, SGGS, p. 837)

Awpy Drql Awip jlu ipAwrw Awpy kry krwieAw]

*The Beloved Himself is the earth, and He Himself is the water;
He Himself acts, and causes others to act.*

(M. IV, SGGS, p. 605)

The next three Chapters discuss Indian Flowers (Lotus in particular) and Spring (*Basant*) with reference to SGGS. I find these Chapters out of tune with the aims and objectives of the book under review. Though I do not find any conflict for inclusion of these Chapters but they are incoherent with the title of this book. Perhaps, the author wants to establish that Indian mythology has been rejected in SGGS: "In Indian mythology, it is stated that though being born from the lotus of Vishnu's navel and having practiced austere penance, Brahma was unable to find the extent of the God's creation. Gurbani describes this fact as:

nwil kutɔbu swiQ vrdwqw bRhmw Bwlx isRsit gieAw]
AwgY AMqu n pwieE qw kw...]

*Brahma, the bestower of blessings, entered the stem of the
lotus, with his relatives, to find the extent of the universe.
Proceeding on, he could not find its limits...*

(M. I, SGGS, p. 350)

The Chapter on Wismad (Wonderful Bliss) has impressed me a lot. Author defines various forms of Wismad and calls it the "The Pinnacle of Spiritual Experience": "We all need to experience *wismad* in order to set a firm foundation for spiritual and psychological development in our lives. In its essence, mystical wismad is spiritual and sacred. We can experience the meaning of life and its much splendour simply by immersing ourselves in this state of wonderful bliss. Gurbani emphasizes that the experience of mystical *wismad* is a sine-qua-non to understand the ultimate reality".

ibsmn ibsm Bey ibsmwdw prm giq pwvih hir ismir prwnl]

*Being wonderstruck with His (Lord's) wonders; meditate on
Him, O mortal, and you shall understand the ultimate reality.*

(M. V, SGGS, p. 893)

The last six Chapters of this volume deserve a special mention as they explore the role which SGGS and Sikhs are destined to play in the present day global crisis. The author is worried about the ecological crisis: "At present, in the midst of a technological revolution, humanity is facing great challenges for its survival. Ecological crisis is one of the gravest among these. There is a serious concern that the earth may no longer be a sustainable bio-system". The chapter "Ecological Concerns in Sri Guru Granth Sahib" discerns the role to be played by SGGS: "SGGS declares that the purpose of human beings is to achieve a blissful state and to be in harmony with the earth and all of God's creation".

"Relevance of Guru Granth Sahib in Twenty First Century" stresses the significance of SGGS in the present century to tackle the problems of Environment, Corruption, Drug Abuse and World Peace. The author quotes profusely from Gurbani to support his reliance and claims on SGGS: "The ideals of the Universal brotherhood of man and the Universal fatherhood of God, as laid down in SGGS are of fundamental importance to settle all conflicts of man":

eyku ipqw eyks ky hm bwirk...]

One True Lord is the father of all; and we are His children...

(M. V, SGGS, p. 611)

The role of Sikhs has been defined in the Chapter "Peaceful Co-existence and Role of Sikhs": "Thus a Sikh's role is to lead an exemplary life based on strong ethics and love for God and all beings. A life-style based on the principles of honest labour (*kirat karo*), sharing with others (*vand chhako*), *Seva* (selfless voluntary service) and *simran* (love of God or meditation) leads to an inner peace in an individual and thus providing a basic framework for fostering peace at family level and in society as well". The role of Sikh organisations is of paramount importance: "Sikh organizations and Institutions should develop a common understanding of the concepts and terms of coexistence".

"Religious Experience in Science" is an important contribution to this volume. The author has supplied the best available quotes from renowned scientists to establish a congruence between Science and Religion. According to author, the study of science helps in the discovery of God: "It has been the experience of many great scientists, that science ultimately leads man to a personal encounter with the absolute or the ultimate reality. An awareness of the ultimate reality may come to the scientist through the perception of order in nature, in the form of a spiritually moving experience". The author has retrieved and collated all the text in this chapter from other sources in a harmonious style.

In the last Chapter "Interpreting Gurbani: Formulating a Methodology", the author explores the topic of Gurbani Hermeneutics from its first principles. He defines the two laws of Gurbani Hermeneutics as follows: " *The first important Law of Gurbani Hermeneutics is that the SGGS should be interpreted literally. Literal Gurbani interpretation means we understand the Gurbani in its normal/plain meaning. The SGGS says what it means and means what it says. The second crucial Law of Gurbani Hermeneutics is that a verse or passage must be interpreted historically, grammatically, and contextually. Historical interpretation refers to understanding the culture, background, and situation which prompted*

the text". I wonder author has relied on one dozen foreign sources for writing about "Various Schools of Gurbani Interpretation" while ignoring completely an equal number of Indian sources. In my view, Tarn Singh, Nirbhai Singh and Gurnek Singh, all from Punjabi University Patiala, must deserve a mention by the author as they wrote their treatises about Gurbani interpretation before and after the era of Hermeneutics.

I am pleased to write this review as a unique opportunity was provided by the author to go through this comprehensive volume of 21 Chapters. I have examined the text critically and find hardly any flaw like typos, misquotes of Gurbani hymns etc., except some lack of references that I have pointed out in my review. I hope the preachers of Sikh religion and exegetes of Gurbani will find this volume as a resource material.